Dineth Wijesinghe

System Engineering Intern

+94 76 911 9747 | dineTH2003-dev | dinethdilshan64@gmail.com | dineth-wijesinghe | portfolio

SUMMARY

Information Technology undergraduate at the University of Moratuwa with hands on project experience administering Linux and Windows environments, configuring cloud virtual machines (AWS), and deploying containerized infrastructure. Practical knowledge of enterprise networking (TCP/IP, DNS, DHCP, VPNs, firewalls), system health monitoring, alert escalation, and automated backup and disaster recovery procedures. Proficient in Bash and PowerShell automation, Docker services, web server administration, and user access management; seeking a System Engineering Internship opportunity to maintain resilient IT infrastructure.

EDUCATION

BSc (Hons) in Information Technology **2024 – Present**

University of Moratuwa, Moratuwa, Sri Lanka

- **GPA: 3.6 / 4.0**

G.C.E. Advanced Level **2020 – 2022**

H/North Kanumuldeniya National School, Hambantota, Sri Lanka

- Combined Mathematics | Physics | ICT · **Z Score: 1.7991**

KEY PROJECTS

UniNet | Cross Platform Network Automation Utility GitHub Repo

▶ **Technologies:** Bash, PowerShell, Linux NetworkManager Dispatcher, macOS launchd, Windows Task Scheduler, curl, GitHub Actions

UniNet is a cross platform, zero external dependency network utility engineered to automatically authenticate campus captive portals seamlessly across Linux, Windows, and macOS environments.

Key features:

- Engineered a cross platform network utility authenticating campus captive portals for Cisco/NEC, Aruba, and Ruijie networks, eliminating repetitive manual browser logins with under 2s reconnect times.
- Implemented event driven hooks via Linux NetworkManager dispatcher, macOS launchd, and Windows Task Scheduler, automating network recovery on state transitions with zero continuous CPU/memory overhead.
- Built cross platform installer and dependency self healing scripts in Bash and PowerShell with automated diagnostic logging, ensuring seamless bootstrap and zero touch troubleshooting.

AWS Three Tier Scalable Web Application Infrastructure GitHub Repo

▶ **Technologies:** AWS (EC2, ALB, Auto Scaling Groups, RDS PostgreSQL, CloudWatch), Terraform, Linux, Docker, Pino, Nginx, GitHub Actions (OIDC)

A production ready, three tier cloud infrastructure provisioned with modular Terraform on AWS, featuring high availability, CPU based auto scaling, and structured CloudWatch monitoring.

Key features:

- Provisioned a Terraform managed three tier architecture across isolated public ALB, private EC2 application, and database subnets with strict Security Group firewalls for network segmentation.
- Configured automated health checks (`/health/ready`), CPU based Auto Scaling, and CloudWatch alarm thresholds, enabling proactive self healing instance replacement and rapid alert escalation.
- Containerized services behind Nginx reverse proxy, configured CloudWatch Agent structured logging, and verified automated PostgreSQL backups, storage quotas, and disaster recovery restoration.

CloudSight | Cloud Infrastructure Observability & Monitoring Platform (Group Project) GitHub Repo

▶ **Technologies:** Linux, Node.js, Express, React, Python, Scikit Learn, Docker Compose, AWS (EC2, RDS, ECR), PostgreSQL, Redis, Nginx, GitHub Actions

CloudSight is a collaborative cloud observability platform featuring real time infrastructure health checks, multi tier incident lifecycle triage, dynamic alerting, and automated container deployment.

Key features:

- Collaboratively engineered an infrastructure monitoring platform with continuous health checks and three tier severity triage (Warning, Critical, Emergency), standardizing escalation to reduce Mean Time to Triage (MTTT).
- Containerized web servers, background workers, Redis cache, and PostgreSQL database with Docker Compose, enforcing restart policies, persistent volume storage, and documented operational runbooks.
- Implemented an automated push to deploy CI/CD pipeline (GitHub Actions + ECR) for zero downtime updates, managing Nginx reverse proxying, Let's Encrypt SSL/TLS, and automated database snapshots.

Tunnix | Self Hosted Ingress & Developer Tunneling Platform GitHub Repo

▶ **Technologies:** Go, gorilla/websocket, Bun, Docker, AWS (EC2, ECS, ECR, EBS, CloudFront, S3), Nginx, Let's Encrypt, SQLite (WAL), TypeScript, Bash

Tunnix is a high performance, self hosted developer ingress platform that securely exposes local HTTP servers to the internet through bidirectional WebSocket multiplexing and automated SSL management.

Key features:

- Developed a high performance Go WebSocket ingress gateway multiplexing concurrent HTTP traffic with under 50ms latency across Linux, macOS, and Windows cross platform CLI agents.
- Containerized the control plane with Docker, managing EBS backed SQLite WAL persistence, automated zero touch Let's Encrypt SSL renewal, and dynamic Nginx reverse proxy configurations via Terraform.
- Hardened infrastructure with a four tier token security hierarchy, sliding window rate limiting, and systemd service supervision scripts, eliminating manual certificate maintenance and downtime toil.

AWS K3s OpenTelemetry Observability Lab GitHub Repo

▶ **Technologies:** AWS EC2, K3s Kubernetes, OpenTelemetry Collector, Honeycomb, OTLP/HTTPS, Docker, Linux, Bash

A production grade Kubernetes observability platform on AWS EC2 running microservices instrumented with OpenTelemetry Collector pipelines and Honeycomb for distributed tracing and triage.

Key features:

- Provisioned a K3s Kubernetes environment on AWS EC2 Linux instances with Security Group network isolation, namespace CPU/memory quotas, and liveness/readiness health probes for automated self healing.
- Engineered a unified OpenTelemetry Collector pipeline exporting system metrics and traces over secure OTLP/HTTPS to Honeycomb, identifying high cardinality p95/p99 latency spikes and service bottlenecks.
- Standardized system troubleshooting procedures and alert dashboards to continuously monitor service availability, accelerating root cause analysis during simulated infrastructure failures.

CERTIFICATIONS & COURSES

- CCNA (Cisco Certified Network Associate) | Enterprise Networking, Routing & Switching, TCP/IP, Subnetting *(Ongoing)*
- Networking Basics | Cisco · Network Architecture, Protocols, IP Addressing & Subnetting
- Linux Unhatched | Cisco Networking Academy · Linux Command Line Interface, Filesystems & Permissions
- LFS101: Introduction to Linux | The Linux Foundation · Linux Administration, Shell Scripting & System Processes
- AWS Cloud Essentials | Cloud Infrastructure, VMs, VPC, IAM, RDS & Security *(Ongoing)*
- LFS158: Introduction to Kubernetes | The Linux Foundation · Architecture, Deployments & Pod Networking *(Ongoing)*

TECHNICAL SKILLS

- **Operating Systems & Administration:** Linux (Ubuntu, Debian, systemd, process/user management, cron, SSH), Windows Server & Desktop, PowerShell, Active Directory / Entra ID, Microsoft 365
- **Networking & Security:** TCP/IP, DNS, DHCP, VPN, Firewalls (UFW, iptables, AWS Security Groups), Subnetting, Routing & Switching (CCNA), Nginx Reverse Proxy, SSL/TLS (Let's Encrypt)
- **Cloud & Virtualization:** AWS (EC2, VPC, ALB, S3, RDS), Docker Containerization, Kubernetes (K3s), Virtual Machines
- **Monitoring, Storage & Backups:** System Health Checks, Issue Alert Escalation, CloudWatch Agent, Prometheus, Storage Quotas, Automated Database Backups (PostgreSQL, SQLite WAL, RDS Snapshots), Disaster Recovery
- **Scripting & Automation:** Bash, PowerShell, Python, Go, SQL, Git, GitHub Actions (CI/CD Pipelines)

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Batch Representative | Faculty of Information Technology, University of Moratuwa** 2026 – Present
- **Programme Committee Member | Riseup Mora 2026, University of Moratuwa** 2026
- **Senior Prefect | H/North Kanumuldeniya National School** 2019 – 2020

REFERENCES

Mr. Indika Wimalasuriya
Principal Director Observability and AIOps
Virtusa (Pvt) Ltd, Colombo, Sri Lanka
Ibwimalasuriya@virtusa.com
+94 71 935 9004

Ms. Nipuni Premadasa
Lecturer (On Contract)
Faculty of Information Technology
University of Moratuwa, Sri Lanka
nipunip@uom.lk
+94 70 170 3862